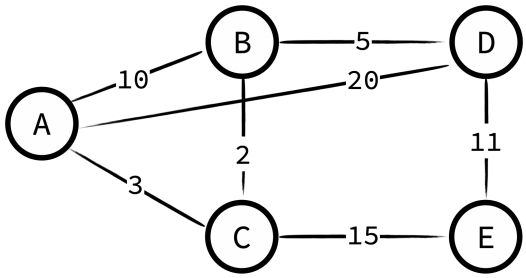
Princess Sumaya University for Technology – King Hussein School for Computing Sciences

CS11313 – **Algorithms Design and Analysis (3)** – **Fall** 2021(1)

Quiz # 4

Name: Solution	**ID:**

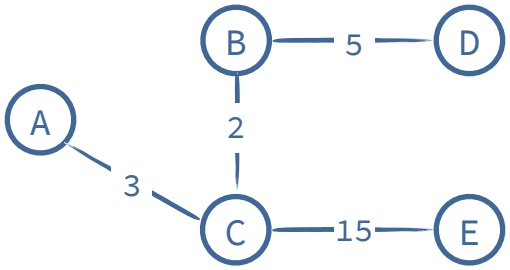


Q1. Assume that **Dijkstra's** algorithm is run on the above graph using vertex A as the source.

(a) List the vertices in the order of their removal from the priority queue (first to last). (3pts)

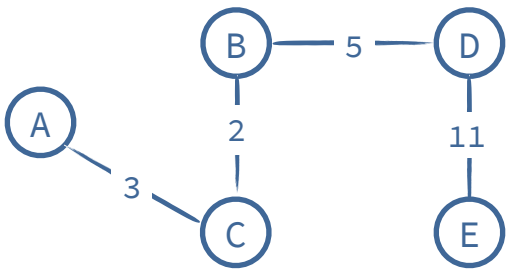
A, C, B, D, E

(b) Draw the resulting shortest paths tree. (3pts



Q1. Assume that **Prim's** algorithm is run on the above graph using vertex A as the source.

(a) List the edges (using their weight) in the order of their insertion into the minimum spanning tree (first to last). (2pts)

3, 2, 5, 11

(b) Draw the resulting minimum spanning tree. (2pts)